

Building an Autonomous Operations Factory

StackGen Introduction

FOUNDING TEAM

Proven team with deep experience at the intersection of AI, DevOps, Infrastructure, SRE, and Security.

FOUNDED IN 2023

Backed by
Thomvest Ventures Seligman
Ventures
(25M capital)

AIDEN products

-  Aiden For InfraOps
-  Aiden For Devops
-  Aiden For Observability
-  Aiden For SRE

KEY CUSTOMERS

 Nielsen

Fortune 5 company

 SIEMENS

 AUTODESK

 LOWE'S

 OneTrust

 CHAMBERLAIN®

 Bancolombia

 INMOBI

Analyst Recognition

- Gartner Cool Vendor - AI in IT Operations, 2025
- Gartner Hype Cycles - Sample Vendor in five 2025 reports: I&O Automation, Platform Engineering, Infrastructure Strategy, Cloud Platform Services, Monitoring and Observability

“

StackGen has been central to how we standardized infrastructure creation across a very complex environment — hundreds of AWS accounts, strict compliance requirements, and a lot of legacy sprawl. It has already helped us cut audience measurement infrastructure costs nearly in half. Bringing delivery, observability and incident management onto the same foundation is the logical next step, and it's where we see the most upside,

 Nielsen

”

“

StackGen turned our infrastructure from script wrangling into building with LEGO blocks. It's been a game changer for our team.

 innovaccer

”

“

The AI SRE agent met all our original goals. We saw the value in first 2 weeks of deployment.

 CHAMBERLAIN®

”

 Tech Mahindra

 cognizant

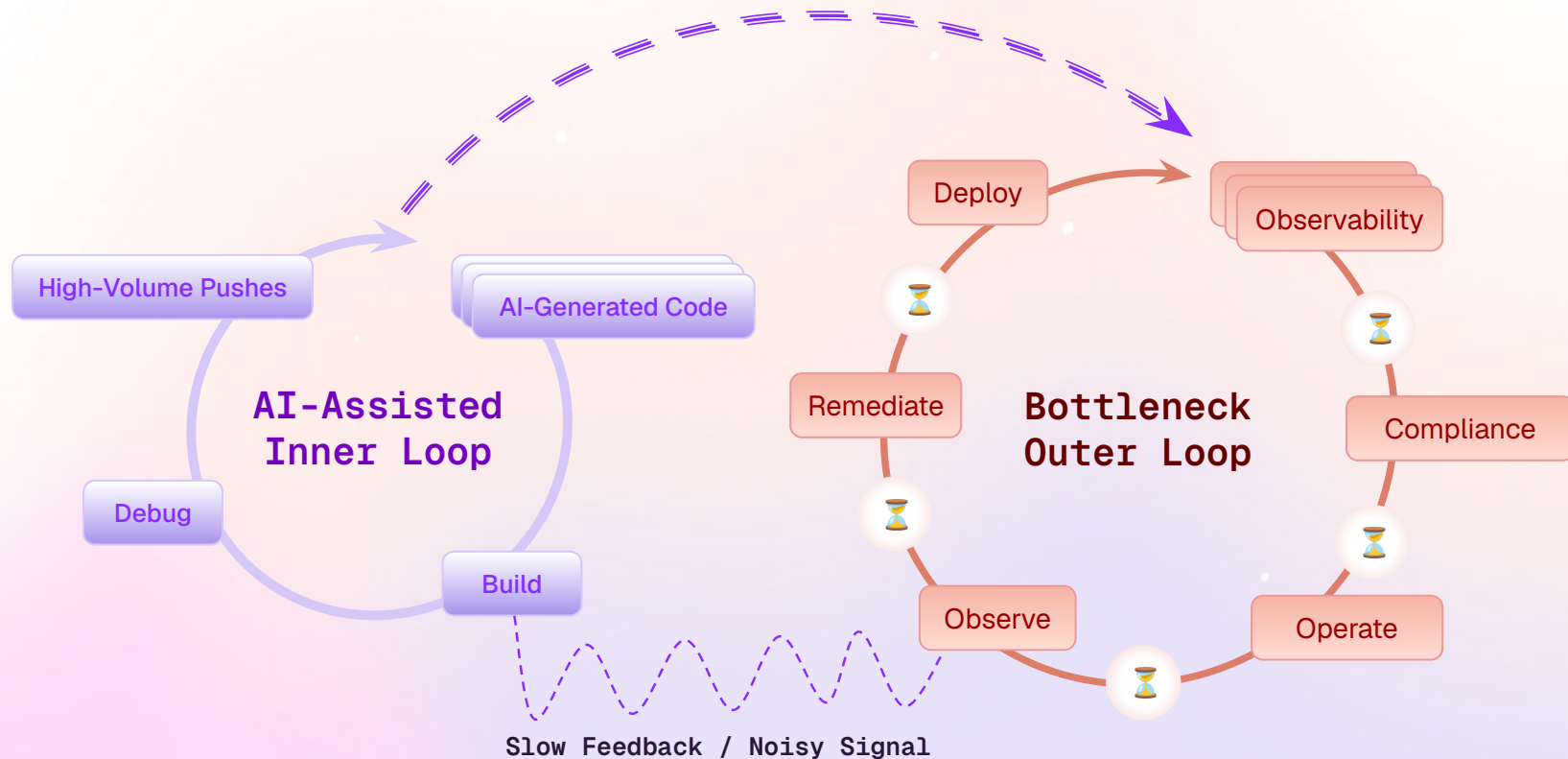
 accenture

 wipro

 Infosys



Outer Ops loop is failing to keep up with inner Dev loop



An increase in AI adoption in software development inversely impacts Delivery Stability.
- Conclusion from DORA, Impact of Generative AI in Software Development, April 2026

Introducing The Autonomous Operations Factory

Outcomes, not Agents.

An Autonomous Operations factory that understands how a company builds, ships, runs and improves software - then turns that context into reliable and repeatable action.



Build



Operate



Observe



Remediate



Learn



Build your Autonomous Operations Factory (short version)

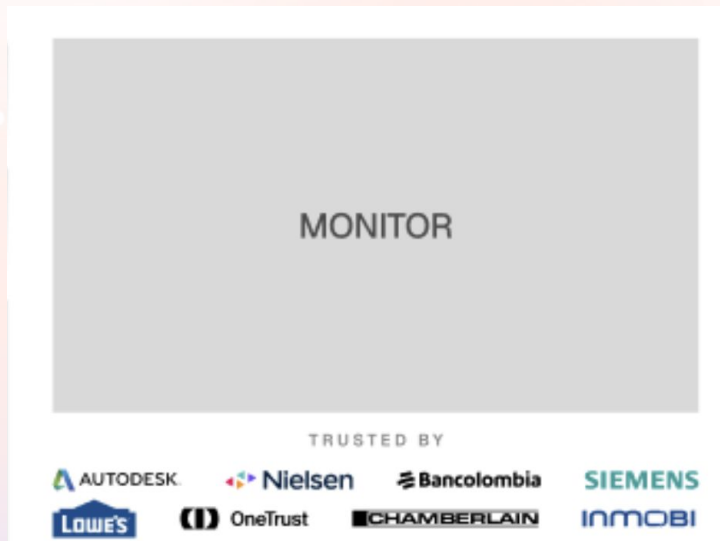
Turn intent into spec-driven,
reliable and auditable cloud

BUILD
100% Guard-railed
Infra

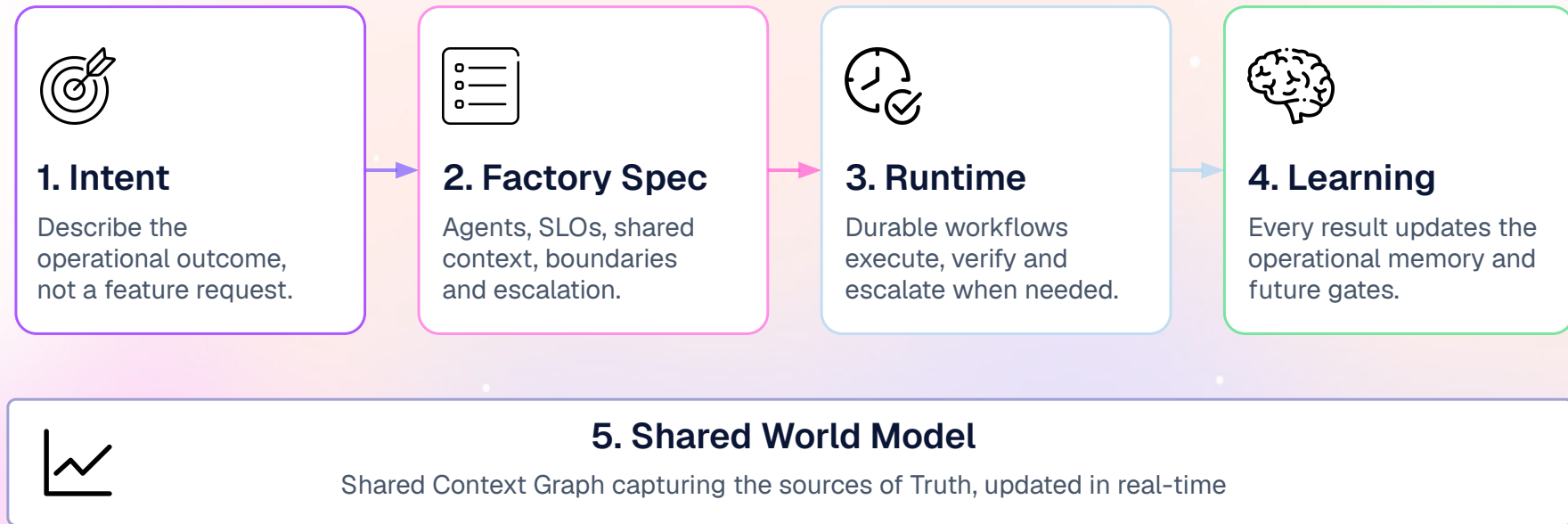
OPERATE
Devops deploys
more frequent

OBSERVE
O11y cost savings

REMEDIATE
SRE toil reduction



Intent to continuous, reliable and auditable execution



What it Does

Agents observe, decide, and act across the SDLC

Intent and Signals

- **Code & PRs**
Commits, reviews, merges
- **CI/CD events**
Builds, tests, deploys
- **Infra state**
Terraform plans, drift
- **Telemetry**
Metrics, logs, traces
- **Incidents**
Alerts, SLO breaches

Core Aiden OS

Active World Model

Code, infra, runtime, and knowledge
— unified, queryable, real-time.

Reasoning Agents

Plan, simulate, and act on grounded
'sources of truth'.

Policy & Guardrails

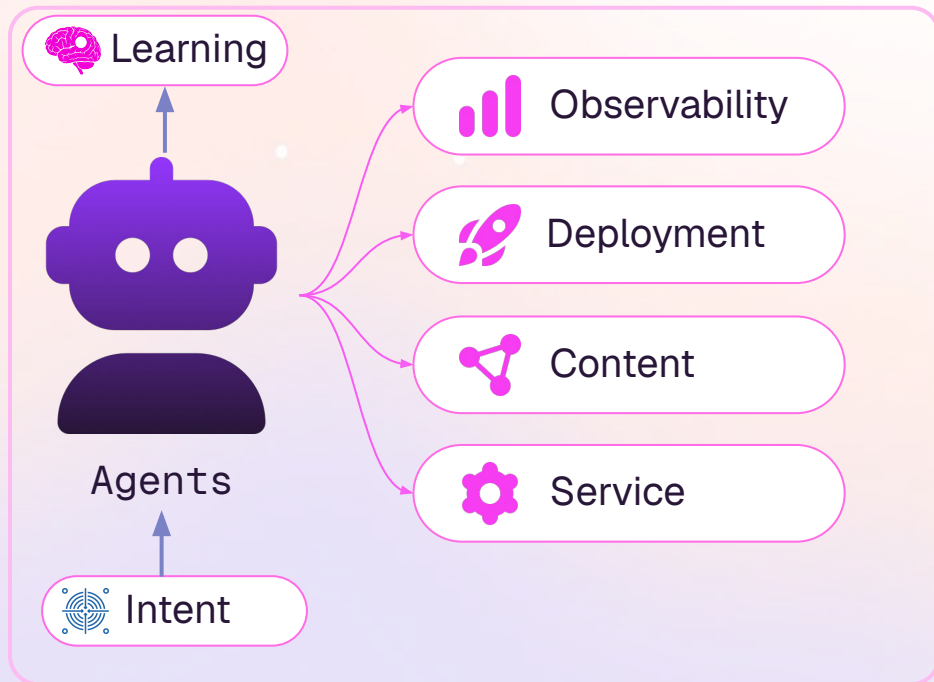
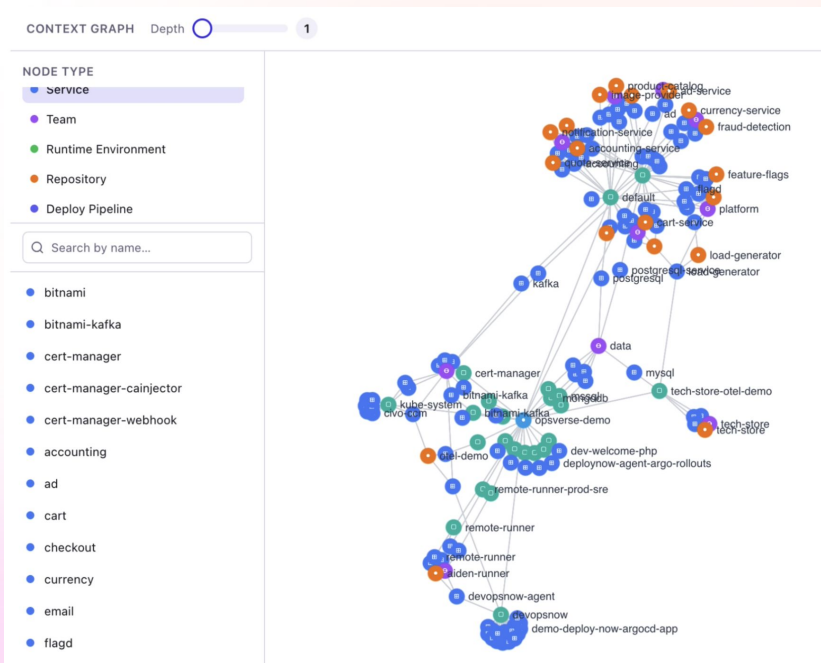
Approvals, blast-radius limits, and
auditable automation.

Agent Actions

- **Open & merge PRs**
Bug fixes, upgrades, refactors
- **Orchestrate pipelines**
Gate, retry, promote, roll forward
- **Remediate drift**
Reconcile infra to intent
- **Defend SLOs**
Throttle, scale, page selectively
- **Safe rollbacks**
Bounded blast-radius reverts

Decision Traces — every input, every decision, every action: durable, replayable, audit-ready.

Aiden World Model - capturing the Four 'Bodies' of Truth



Humans keep authority. The factory absorbs toil.

Humans own

- Goals and risk appetite
- Customer promises and trade-offs
- Novel exceptions and judgment
- Authority for irreversible impact

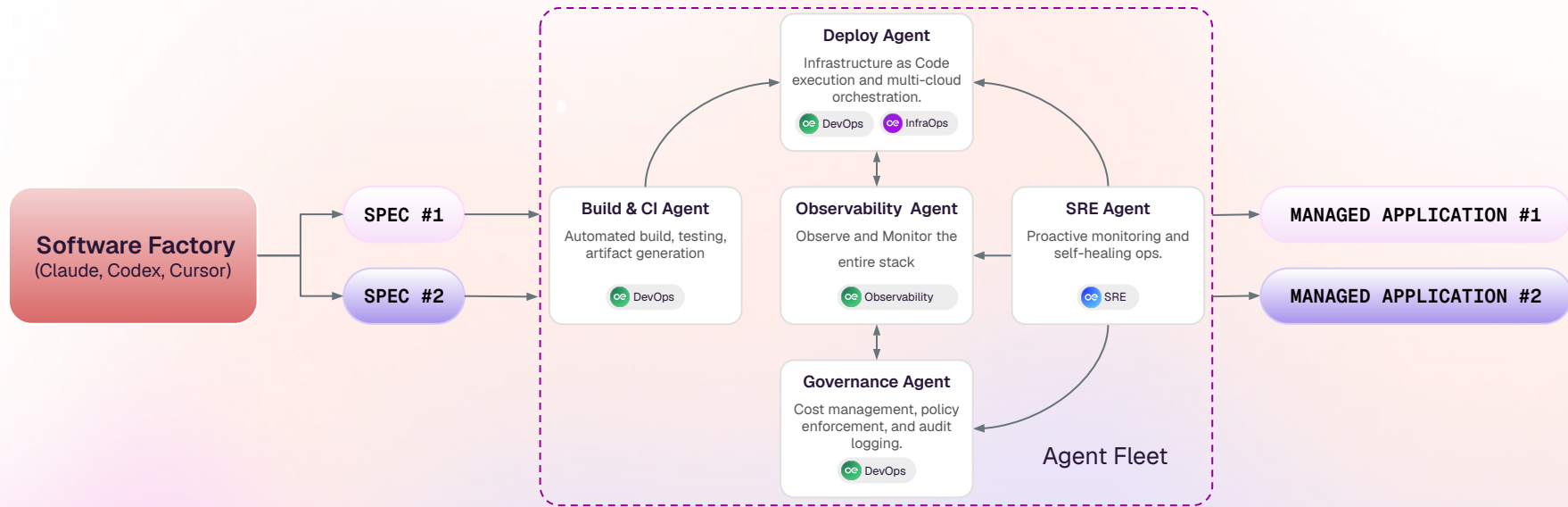
Factory owns

- Evidence gathering and correlation
- Routine bounded decisions
- Execution, verification and rollback
- Reusable learning from every outcome

One Command Center* for operations

**Command center currently in preview*

The Autonomous Operations Factory - How it works



Aiden OS

Agentic Harness for reliable, governed, auditable execution



HUMAN-IN-THE
LOOP APPROVAL



DETERMINISTIC
EXECUTION



World Model



APP FRAMEWORK



GOVERNANCE +
RBAC



POLICIES +
AUDITABILITY

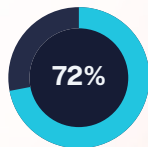


MEMORY



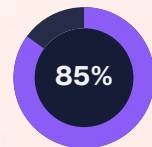
TOKENOMICS

Autonomy Index - measuring Factory Autonomy and Flow



Autonomy Index

% of factory decisions (merge, release, rollback, remediation) resolved by agents without human escalation, weighted by action risk and complexity.



Autonomy Coverage

% of the value stream (plan → build → test → deploy → operate) running on agent-executed pipelines.

BY LINE OF BUSINESS



Payments



Digital Channels



Data Platform



Corporate IT

BY TEAM



Core Services



Mobile Apps



Platform / SRE



Integrations

BY AGENT TYPE — THE FLEET FROM THE PREVIOUS SLIDE



Builder



Quality



Release



SRE



Governance



Orchestrator

Donut = Autonomy Index | bar beneath = Autonomy Coverage. Values are illustrative target-state benchmarks — no external industry baseline exists for these metrics yet.

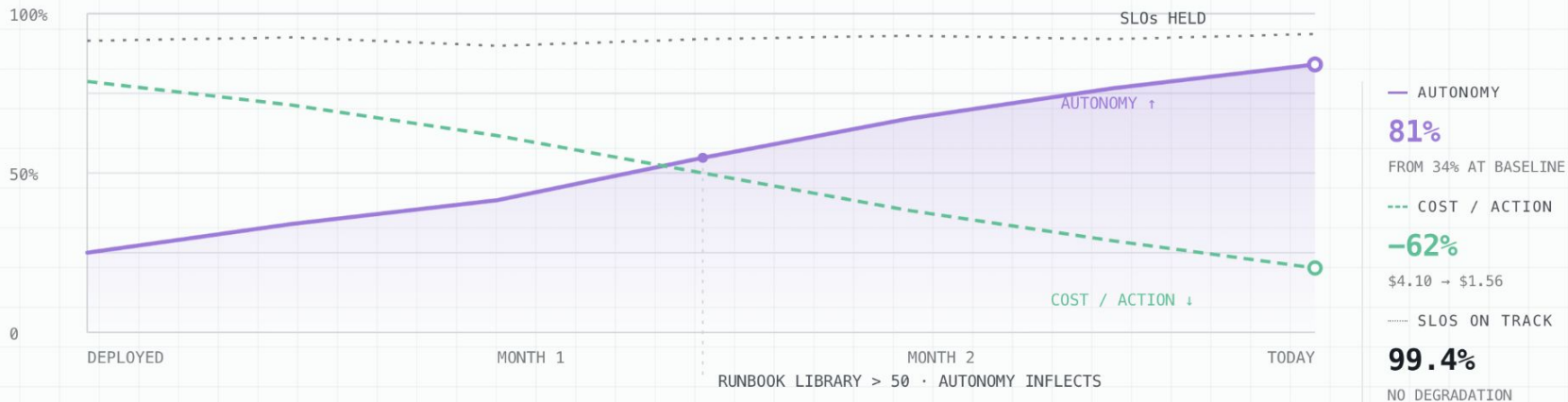
DORA metrics, reframed for an agent-run factory



The factory model closes gaps in DORA metrics by measuring agent Autonomy across for full-lifecycle SLOs, not just functional silos.

Aiden now handles **81% of operations autonomously**, at **62% lower cost per action** than at deployment, with service levels **holding at 99.4%.**

ACME CORP · 1 WORKSPACE · 2 APPS DEPLOYED · BASELINE SET 2026-05-04



METHODOLOGY | Every figure derives from logged agent actions and your own baseline. [View calculation basis →](#)

02 OPERATING DIMENSIONS

each drills to detail

AUTONOMY

81%

of actions need no human

BASELINE 34%

BY WORKSPACE →

COST

-62%

cost per action vs baseline

\$1.56 / ACTION

COST DETAIL →

RELIABILITY

99.4%

SLOs on track · 3 breaches

1,284 INCIDENTS

GOVERNANCE →

SPEED

-48%

time to production

LEAD TIME 3.2D

PIPELINE →

Current Offering



Aiden for InfraOps

Infrastructure
Lifecycle
Management



Aiden for DevOps

DevOps workflows



Aiden for Observability

Observability



Aiden for SRE

Reliability &
Remediation

Customer Apps

App

App

App

App

App

App

Aiden OS

Agentic Harness for reliable,
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HUMAN-IN-THE
LOOP APPROVAL



DETERMINISTIC
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World Model



APP FRAMEWORK



GOVERNANCE +
RBAC



POLICIES +
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MEMORY



TOKENOMICS

DEMO

Demo #1: Migration Factory Powered by StackGen

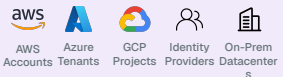
Acquisition Sources

Unknown & Undamaged
Acquired Environments

Acquired Company A



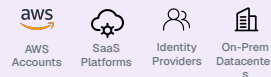
Acquired Company B



Acquired Company C



Future Acquisitions



StackGen now is achieving critical artifacts, your critical artifacts. Resource reports.

Cloud Discovery Engine

- Resource Inventory
- Cloud Asset Discovery
- Service Enumeration
- Configuration Analysis

Architecture Intelligence

- Dependency Mapping
- Application Topology
- Service Relationships
- Network Visualization

Cloud-to-Code Engine

- Terraform & OpenTofu
- Infrastructure Codification
- State Reconstruction

AI Operations Layer

- Drift Detection
- Compliance Validation
- Policy Enforcement
- Remediation Recommendations

StackGen generations:

- Resource graphs
- Dependency Maps
- Architecture Diagrams
- Terraform Code
- OpenTofu Code
- Compliance Reports

home-grown platform orchestrates the entire acquisition lifecycle.

STACKGEN INTELLIGENCE LAYER

Acquisition Intake

Automated Assessment

Environment Classification

Migration Planning

Landing Planning

Landing Zone Assignment

Modernization Pipeline

Manages Service Handoff

home grown platform orchestrations:

- Discovery Results
- Migration workflows
- Risk assessment
- Resource Classification
- Ownership Mapping
- Application Rationalization

Validates all StackGen-produced content before deployment

GOVERNANCE FRAMEWORK

Identity Governance

- Azure Entry ID
- B2B Federation
- RBAC
- PIM

Security Governance

- Encryption Standards
- Network Controls
- Logging Standards
- Security Baselines

Operational Governance

- Monitoring
- Backup
- Incident Management
- Ownership Tracking

Cost Governance

- Chargeback
- Budget Control
- FinOps
- Resource Optimization

Policy-as-Code

- Azure Policy
- Terraform Guardrails
- Compliance Automation
- Drift Prevention

Transitional Landing Zone

Trusted M&A Landing Zone

POLICY ENFORCEMENT GATES



Management Groups



Subscriptions



Resource Groups



Virtual Networks



Identity Controls



Shared Devices



Monitoring



Security Tooling

Target State Platforms

1. CUSTOMER MANAGES SERVICES

Azure managed platforms

Corporate platforms

Production landing zones

Enterprise shared services

2. APPROVED EXTERNAL PLATFORMS

EXT Workloads

Transitional workloads

Long-term isolated platforms

DEVELOPER SELF-SERVICE

Developer

AI Request

Golden Path Templates

Policy Validation

Terraform Generation

Git Repository

CI/CD Pipeline

Deployment

Self-service Infrastructure with Enterprise Guardrails

Measurable business & operational

Faster Acquisition Onboarding

Automated Environment Discovery

Reduced Manual Terraform Creation

Standardized Landing Zones

Consistent Security Enforcement

Accelerated Cloud Migrations

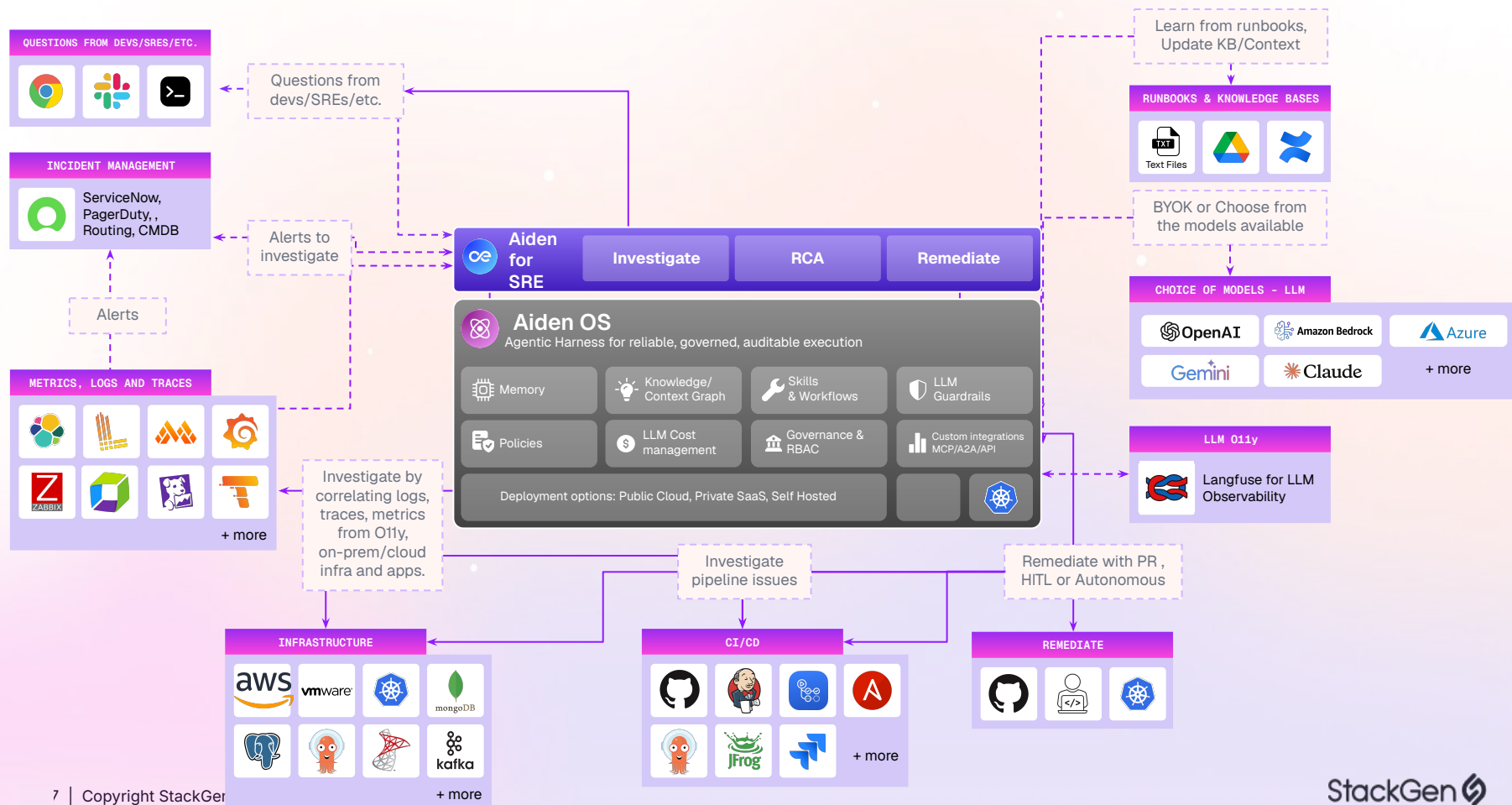
Reduced Configuration Drifts

Faster Managed Service Readiness

Improved Compliance Posture

Scalable M&A Integration Model

Demo #2: How Aiden for SRE Works



Thank you!

BACKUP

How do you get started?

The Path to Autonomous Operations

- **Integrate repo, pipeline, ticket system, observability etc :** Integrate all your DevOps tools. Factory calculates your Autonomy Index and Coverage and makes recommendations for improving your software delivery operations.
- **Build the Knowledge and Context Graphs:** Integrate the 'Four Bodies' into a queryable Knowledge Graph which can deliver the Aiden World Model. Agents come next.
- **Embed agents in the path to Production:** Deploy Agents to correlate Code, IaC, Telemetry signals and memory continuously - not after the incident.

The Autonomous Operations Factory



1. BUILD IT RIGHT

Every rule — security, cost, quality — written as code and enforced automatically.



2. PROVE IT COMPLIANT

Compliance and audit evidence, generated continuously — provable, not a scramble.



3. RUN IT SAFELY

Production detects and heals issues on its own — within limits of the security controls and cost guardrails.

One spec

Shared World Model

The shared source of truth, from first commit to self-healing production

NEEDS YOU 4 OPEN OLDEST 2H AGO	HANDLED AUTONOMOUSLY 92% 287 OF 312 SIGNALS	MEDIAN RESOLUTION 4.2 MIN WAS 6.1MIN	OVERRIDES 3 1.2% OF 247 AGENT ACTIONS	LEARNINGS PENDING 2 / 7 NEW AWAITING APPROVAL
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01 48-HOUR SNAPSHOT

vs prior window ACTIVITY all apps

INCIDENTS & ALERTS resolved 331 resolved ▲ 22% vs prior 147 INCIDENTS · 184 ALERTS	DEVOPS TICKETS triaged 96 tickets ▲ 18% vs prior 91% AUTO-CLASSIFIED	RESOURCES provisioned 34 resources ▲ 12% vs prior 0 DRIFT ON NEW	RUNBOOK MATCHES match rate 342 matches ▲ 18% vs prior 87% AVG CONFIDENCE · 27 BELOW GATE
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Drift detected on payment DB
 Manual change to performance_insights_enabled flagged for review.
 JUST NOW SRE

Runbook update suggested
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 2 MIN AGO SRE

02 WHAT YOUR TEAM DIRECTED

guardrails, policy, judgement

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oldest 2h ago

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SRE · ECS-PAYMENT-SERVICE

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D
Runbook match below threshold
MED

DEVOPS · LIN-4021

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Confidence **71%**

Incident remediated
 checkout-service 500 errors resolved by scaling DB connections.
 42 MIN AGO SRE

Scheduled import complete
 31 tickets ingested from Jira and Linear into the DevOps inbox

Aiden for SRE

For OSS & managed Observability Stack

Aiden For SRE Delivers Immediate Value

INTEGRATIONS: Out of the box integration with OSS version of Grafana, Grafana Cloud, New Relic, Dynatrace, DataDog, OTEL, Kubernetes, Terraform, Git, ServiceNow, Cloud accounts



ALERT INTELLIGENCE

AI reduces alert noise by clustering signals and identifying meaningful incidents.

DETECT



INCIDENT TRIAGE

AI correlates alerts, metrics, logs, and traces to identify the most impactful incidents first.

TRIAGE



RCA ACCELERATION

AI accelerates analysis of telemetry, deploy events, and system context to identify likely root causes.

DIAGNOSE



AUTONOMOUS REMEDIATION

AI recommends or executes fixes within policy guardrails

REMEDIATE

Aiden for SRE

Less toil. Improve Reliability.

- Triage 100s of daily alerts down to the handful that matter; automatically
- Four-step flow: triage, SLO prioritization, RCA, human-approved remediation
- Integrations: Your Stack
- Never acts without sign-off; full audit trail always

50%

Reduction in
MTTR

50%

Reduction in
False Positives

40%

Auto-Remediated
Incidents

35%

Reduction in SLO
Violations

SRE Technical Leader #1

On Aiden's versatility

"We're doing something with Aiden for incident. Writing playbooks we are still leveraging the same Aiden."

SRE Technical Leader #2

On RCA Reduction

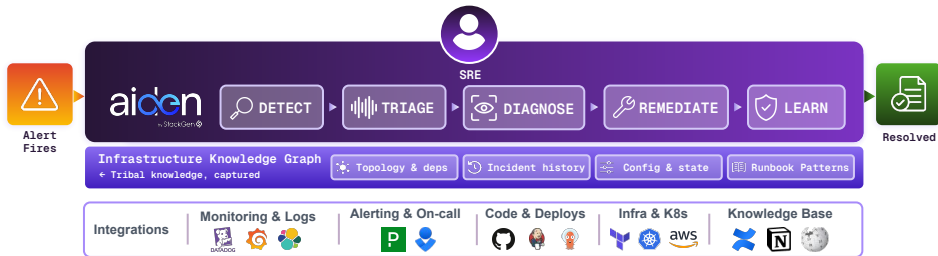
"Before Aiden, forming a hypothesis took 10–15 min. With Aiden, we identified a CrashLoopBackOff in <2 min — **80% reduction.**"

One Platform. Every Signal. Every Team.

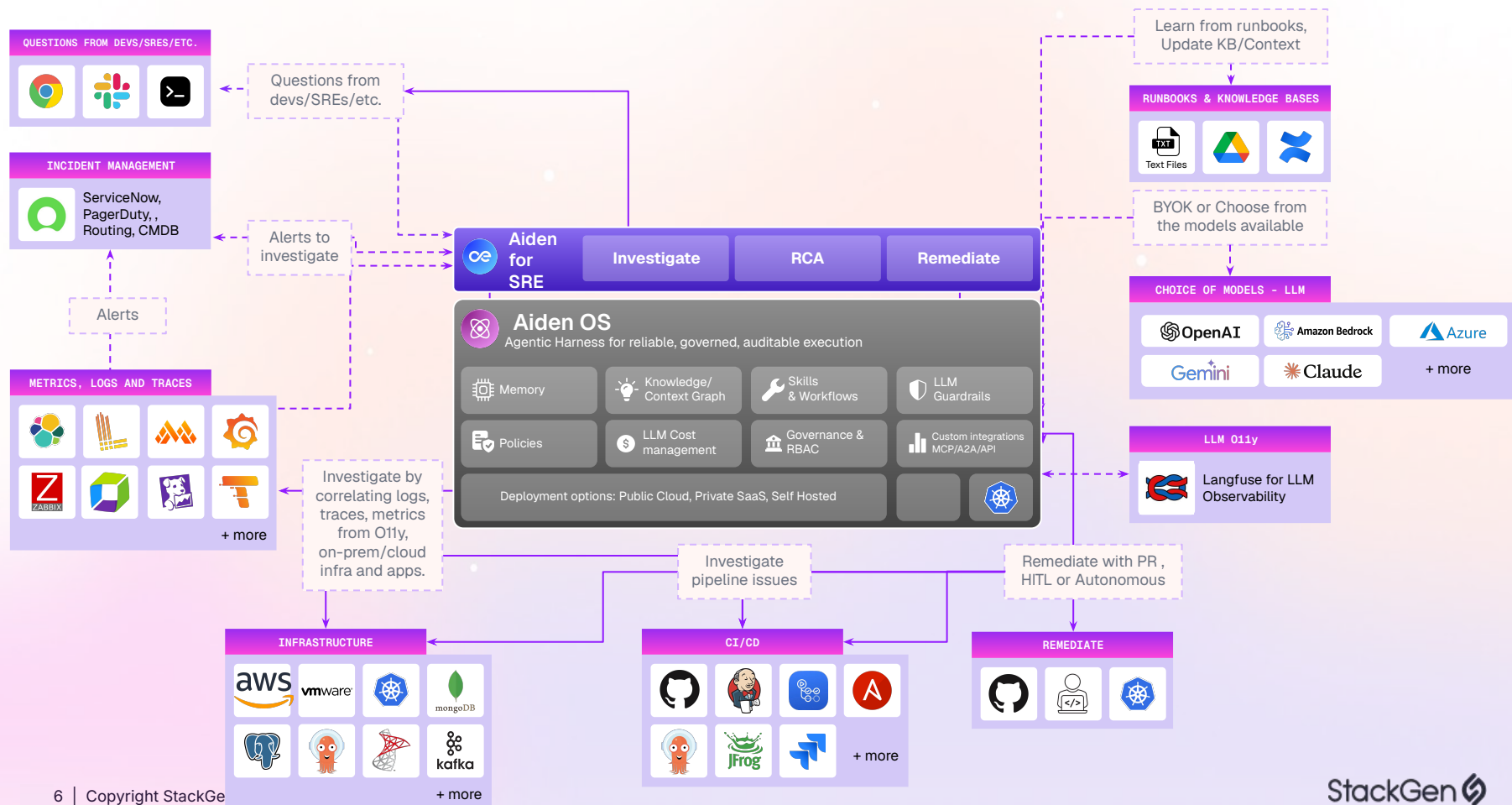
OLD WAY: HOURS TO RESOLVE



NEW WAY: MINUTES TO RESOLVE

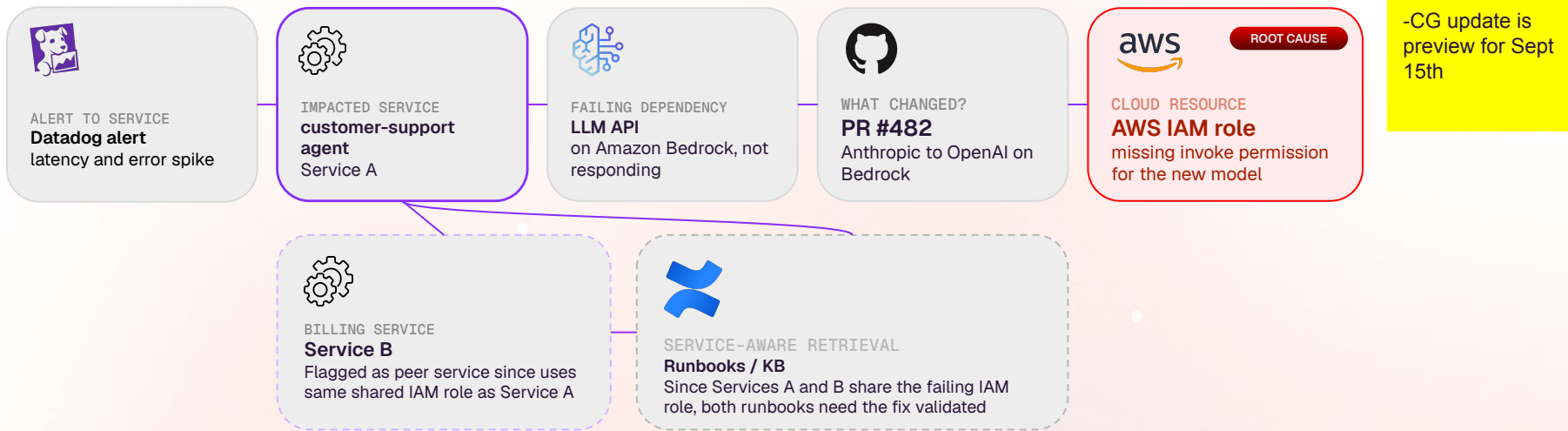


How Aiden for SRE Works



USE CASE

Root Cause Analysis



OPERATIONAL CONTEXT GRAPH



Services



Alerts



Runbooks



Repos



Cloud Inventory

Pull-based 30-day backfill across git, Datadog, AWS account and Terraform deploy, so prior history is available, not just new events.



Aiden

customer-support agent's IAM role is missing Bedrock invoke permission for the OpenAI model introduced by PR #482. Recommended next step: restore the invoke permission, or revert the LLM provider change.



Datadog alert



Blast radius: Services A and B



Confluence runbooks



PR #482



AWS IAM policy



30-day history

Aiden for InfraOps

- Self-service of standardized infra from IDE
- Cloud migration factory

Intent/prompt to governed Infrastructure from IDE

A developer describes the intent to build an AI agent for financial services in Kiro. **Aiden for Infra Kiro Power** turns that intent into compliant, production-ready infrastructure, with no platform-team handoff.



1/5

INTENT · KIRO

Describe Intent

"Support agent for financial services," needing Amazon Relational Database Service (Amazon RDS), Amazon ElastiCache, Amazon Bedrock, AWS App Runner.



2/5

DETECT AND GENERATE

Aiden writes the IaC

Detects the intent, generates secure Terraform / OpenTofu, without leaving the integrated development environment (IDE).



3/5

GOVERN · PRE-PROD

Compliance enforced first

PCI-DSS, SOC2, ISO27001, CIS checked against shared state before production.



4/5

DEPLOY · AWS

Governed resources ship

Deployable AWS resources provisioned in one flow. Drift and misconfiguration caught early.



5/5

CLOSE THE LOOP

ServiceNow audit trail

Auto-files a ticket with pull request (PR) details and deployment-run artifacts attached.

MANUAL HANDOFF

Eliminated

COMPLIANCE

By default

AUDIT TRAIL

Auto-filed

MULTI-CLOUD

IaC for AWS/Azure/GCP

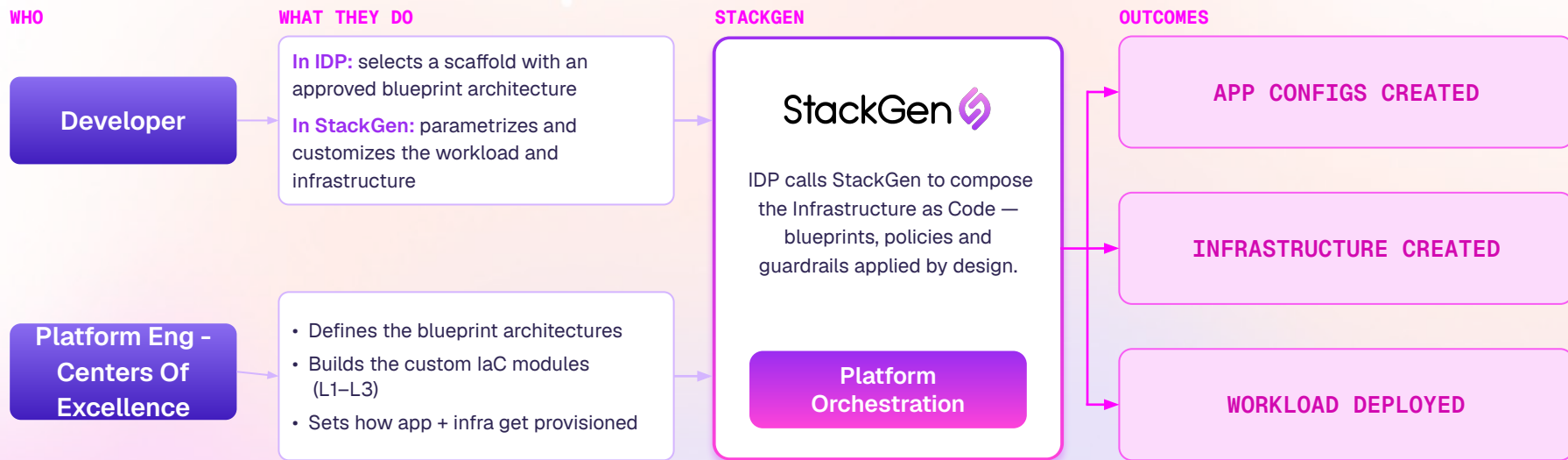
Aiden for DevOps

- From IDP / ticketing system to Automated pipeline provisioning

Developers self-serve from in-house IDP / ticketing systems

StackGen composes, enforces your blueprints, and provisions the stack.

syncAiden integrates out-of-the-box with **ServiceNow & Jira**



From ~6 weeks to days.

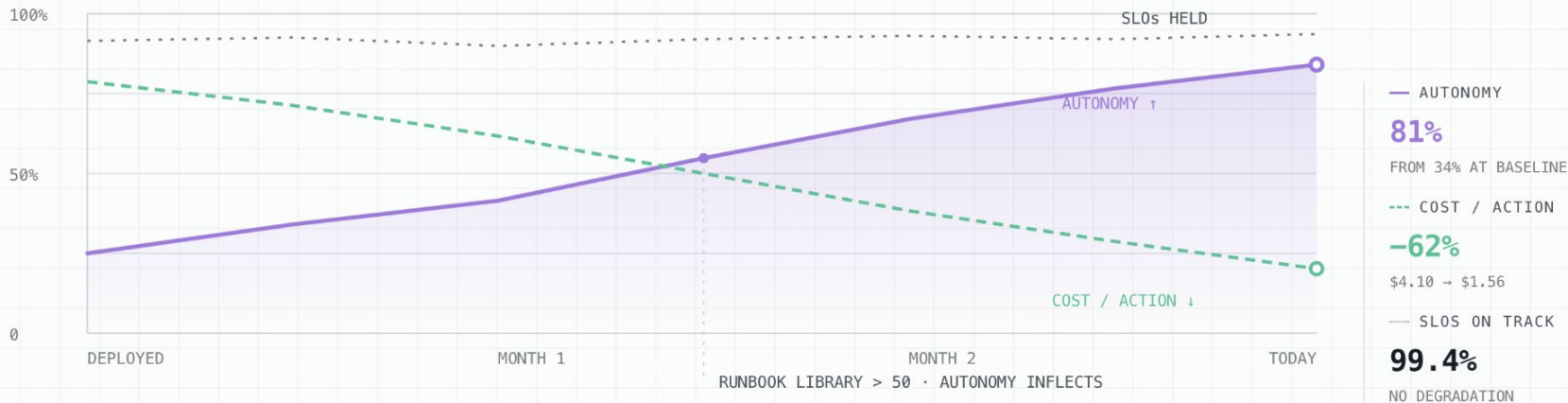
Same governance, same blueprints, same security guidelines — enforced automatically instead of reviewed manually.

Autonomous Operations Factory

- Command center [currently in preview]

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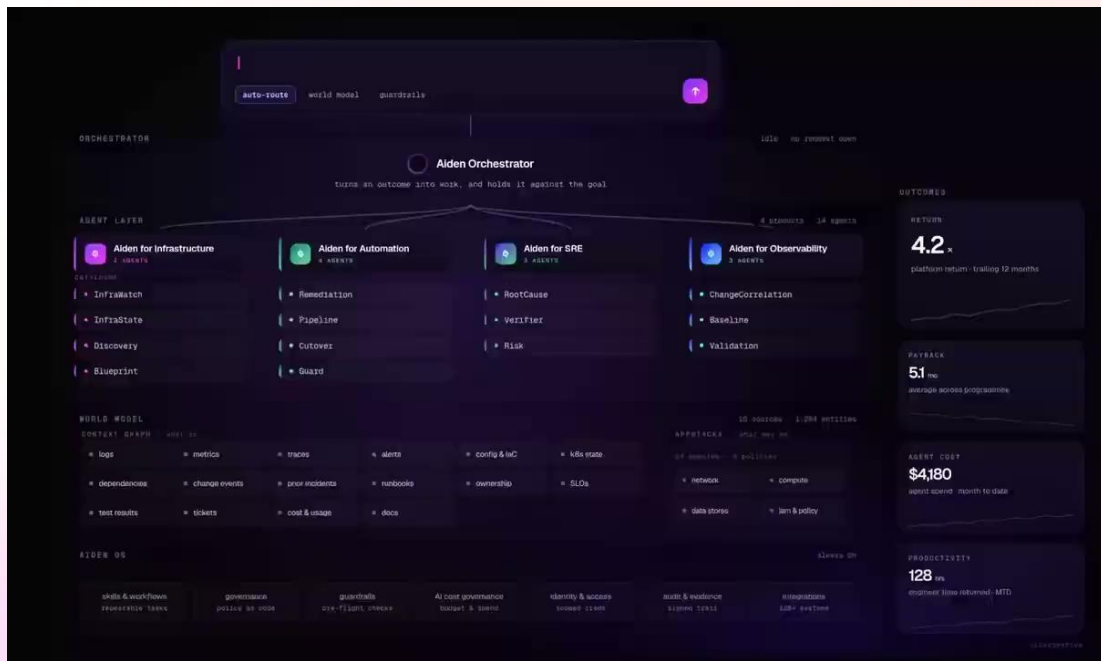
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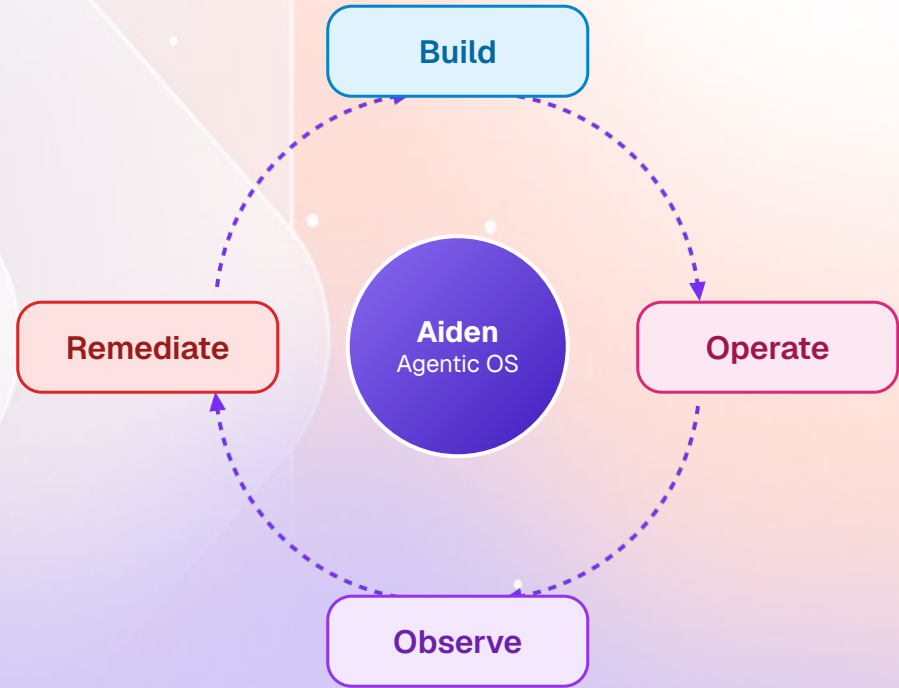
Autonomous Operations Factory

Companion to your Software Factory

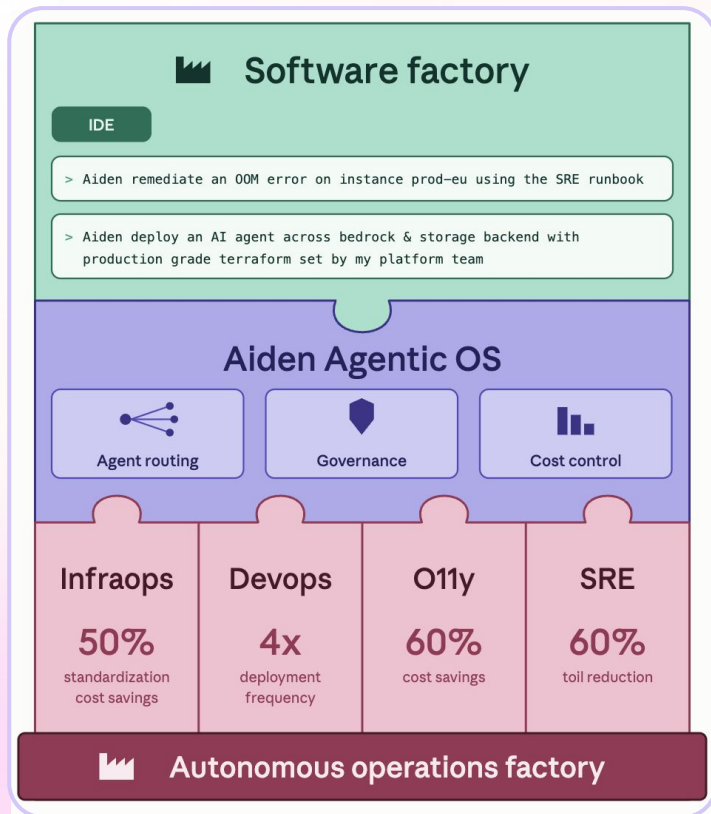


Journeys→

Aiden breaks the silos within your software factory.



Aiden across your Software Factory



1. Software Factory: Aiden is designed to complement the software factories companies have already built, seamlessly integrating with agentic IDEs & enterprise IDPs for production.

2. InfraOps: Infrastructure & base image served with standardized Terraform / OpenTofu Appstacks provides cost savings

3. DevOps: More deployments across dev, test, sandbox, and production environments.

4. Observability: AI agent provides developer time savings by working with existing Grafana, Datadog, New Relic, Dynatrace dashboards.

5. SRE: Toil reduction with repetitive alert triaging & incident remediation.

6. Agentic OS: Intelligent model layer & context graph that works with data from across infra, pipelines, observability, and SRE knowledge base. Agent layer provides routing to the right domain agents;

Aiden agents have inbuilt governance with OPA policies, audit layer, and cost aware model routing. Lower Agentic layer administration burden.